Year 8
Mathematics

Unit 4: 2D Shapes and 3D Solids

4.3 Properties of 3D Solids

Properties of 3D Solids

STARTER · 5 MIN

Date: _____ **Title: Properties of 3D Solids**

LEARNING OBJECTIVES

- Identify nets of different 3D solids

STARTER *Answer in your book. No calculators.*

1. How many faces does a cube have?
2. Name a 2D shape with 3 sides
3. How many sides does a pentagon have?
4. What shape is each face of a cube?

Starter — Answers

STARTER

Tick or correct in green pen

1. **6**
2. **Triangle**
3. **5**
4. **Square**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can match 3D solids to their nets	☐
SILVER	I can identify the edges, faces and vertices of 3D solids	☐
GOLD	I can sketch nets of 3D solids	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Properties of 3D Solids

INTRODUCTION · 3 MIN

A net is a 2D shape that folds up to make a 3D solid.

Face

A flat surface.

A cube has 6 faces

Edge

Where two faces meet.

A cube has 12 edges

Vertex

A corner (plural: vertices).

A cube has 8 vertices

Net

All the faces laid out flat.

It folds to make the solid

Examples 1 and 2 — Faces, Edges and Vertices

EXAMPLES · 5 MIN

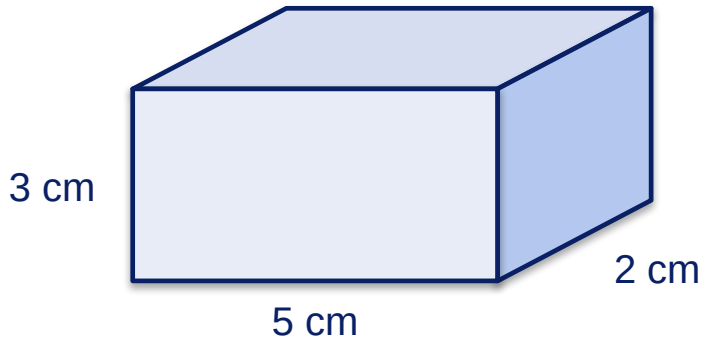
BRONZE

Watch and copy into your book

Hint Face: flat surface Vertex: a corner

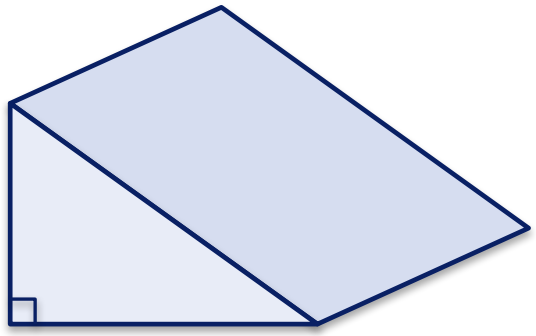
Example 1

Count the faces, edges and vertices of this cuboid.



Example 2

Count the faces, edges and vertices of this triangular prism.



Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Count each one carefully

Faces = 6

Edges = 12

Vertices = 8

Tip: count the hidden ones at the back too.

Example 2

2 triangles + 3 rectangles

Faces = 5

Edges = 9

Vertices = 6

Tip: each triangle has 3 corners.

Task 1

PRACTICE · 9 MIN

BRONZE *Write down the number*

Hint Face: flat surface Vertex: a corner

1. Faces of a square-based pyramid
2. Edges of a square-based pyramid
3. Vertices of a triangular prism
4. Edges of a cube
5. Vertices of a square-based pyramid
★ Faces of a pentagonal prism

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1 **5**

2 **8**

3 **6**

4 **12**

5 **5**

★ **7**

Examples 3 and 4 — Matching Nets

EXAMPLES · 5 MIN

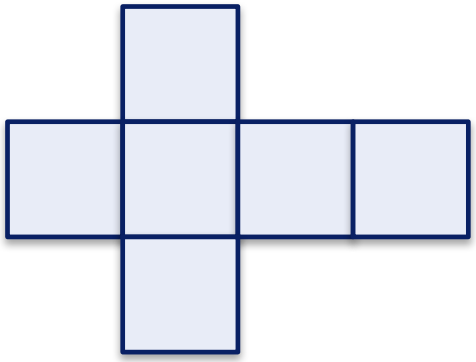
SILVER

Watch and copy into your book

Hint Count the faces in the net Imagine folding it

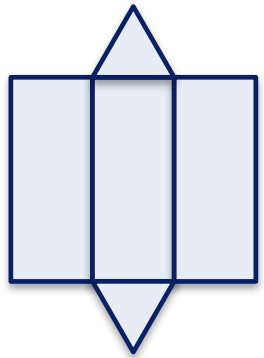
Example 3

Name the solid.



Example 4

Name the solid.



Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

6 equal squares

It makes a cube.

Tip: imagine folding up the 4 sides.

Example 4

2 triangles and 3 rectangles

It makes a triangular prism.

Tip: the triangles are the two ends.

Task 2

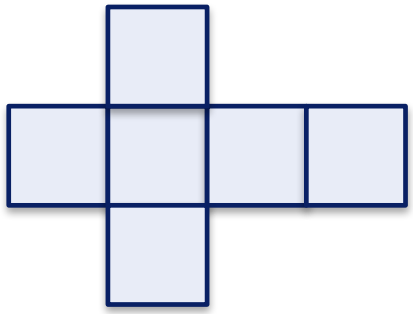
PRACTICE · 9 MIN

SILVER

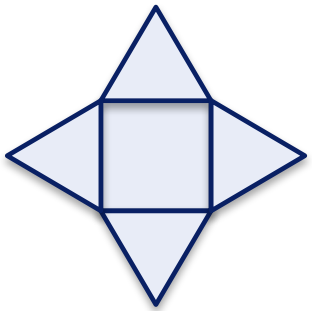
Which solid does each net make?

Hint Count the faces in the net Imagine folding it

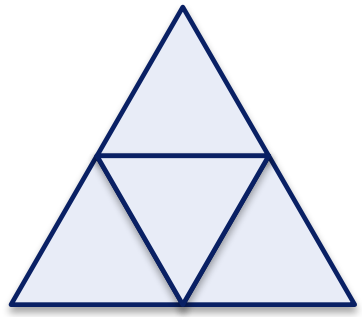
1 Name the solid



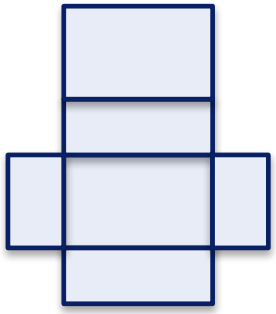
2 Name the solid



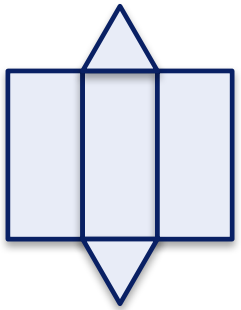
3 Name the solid



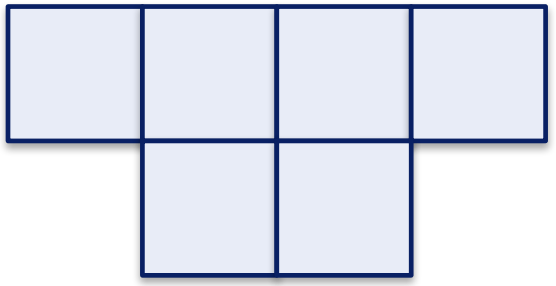
4 Name the solid



5 Name the solid



★ Does it fold to a cube?



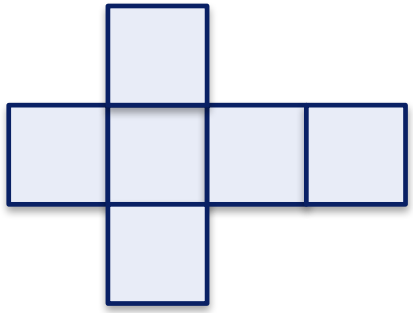
Task 2 — Answers

CORRECTIONS

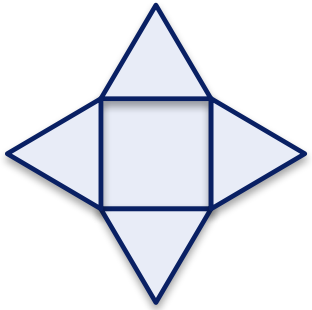
SILVER

Tick or correct in green pen

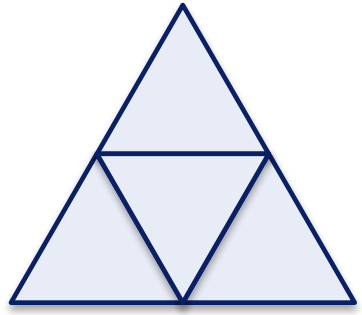
1 **Cube**



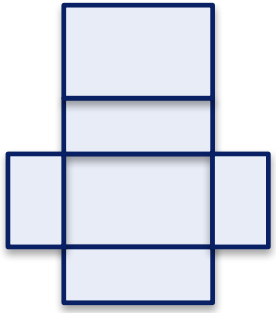
2 **Square-based pyramid**



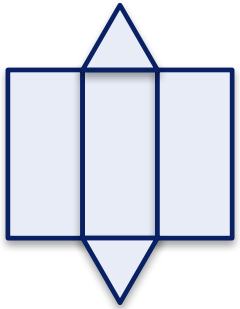
3 **Triangular-based pyramid**



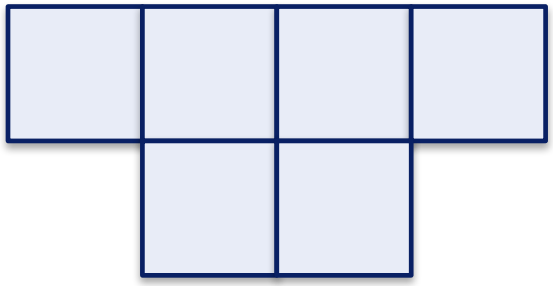
4 **Cuboid**



5 **Triangular prism**



★ **No: faces overlap**



Examples 5 and 6 — Sketching Nets

EXAMPLES · 5 MIN

GOLD

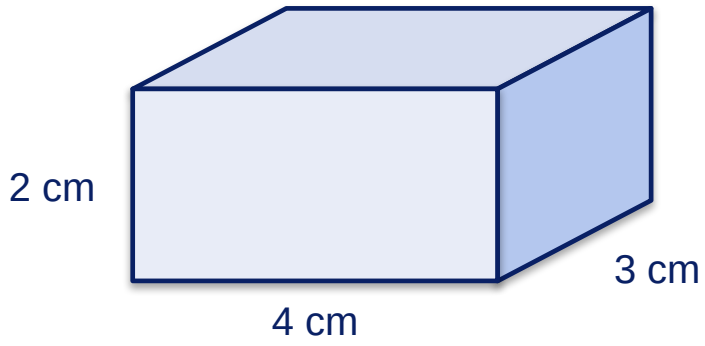
Exam-style: show all your working

Hint Opposite faces are equal F + V − E = 2

Example 5

(2 marks)

Sketch a net of this cuboid.



Example 6

(3 marks)

A solid has 2 triangle faces and 3 rectangle faces.

Name it. Count its edges and vertices.

Examples 5 and 6 — Corrections

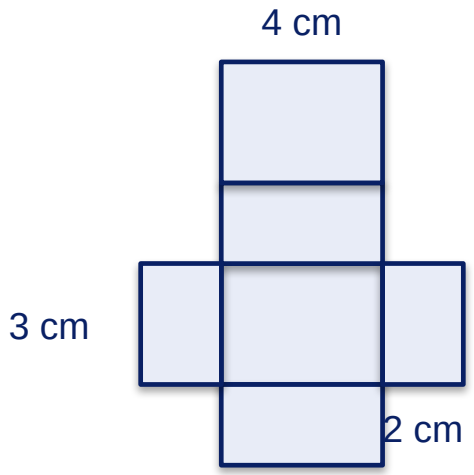
CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

6 rectangles, opposite faces equal



Tip: two 4 by 3, two 4 by 2, two 3 by 2.

Example 6

Ends are triangles

Solid = triangular prism

Edges = 9

Vertices = 6

Tip: check: 5 + 6 − 9 = 2.

Task 3

PRACTICE · 9 MIN

GOLD *Exam-style: solids and nets*

Hint Opposite faces are equal F + V − E = 2

1. Solid with 6 square faces

2. Solid with 1 square face and 4 triangle faces

3. Number of faces in a cuboid net

4. Cube: faces + vertices − edges

5. Triangular prism: faces + vertices − edges

★ A prism has 8 faces. How many sides has its end?

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. **Cube**

2. **Square-based pyramid**

3. **6**

4. **6 + 8 − 12 = 2**

5. **5 + 6 − 9 = 2**

★ **6 (hexagonal prism)**

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE How many faces does a cube have?

SILVER What is the difference between a face and an edge?

GOLD Which solid has 2 triangle and 3 rectangle faces?

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **6**

SILVER **Face: flat surface. Edge: where two faces meet**

GOLD **Triangular prism**

Exam Practice — Faces, Edges and Vertices

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

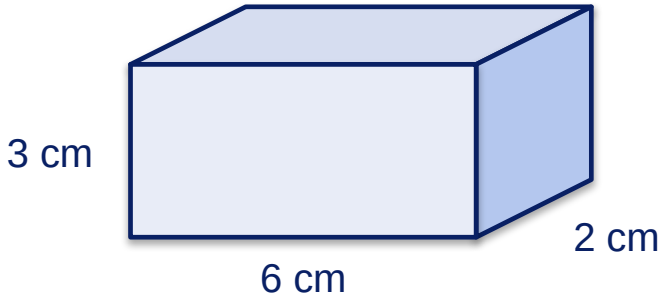
Hint Face: flat surface Vertex: a corner

Question 1

(3 marks)

Here is a cuboid.

Diagram NOT accurately drawn



(a) How many faces? (1)

.....

(b) How many edges and vertices? (2)

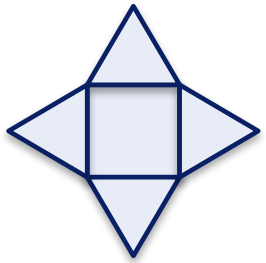
.....

Question 2

(2 marks)

Here is the net of a solid.

Diagram NOT accurately drawn



(a) Name the solid. (1)

.....

(b) How many vertices? (1)

.....

Exam Practice — Nets

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

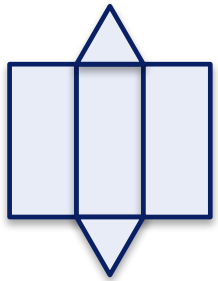
Hint
Count the faces in the net
Imagine folding it

Question 3

(2 marks)

Here is the net of a solid.

Diagram NOT accurately drawn



(a) Name the solid. (1)

.....

(b) How many edges? (1)

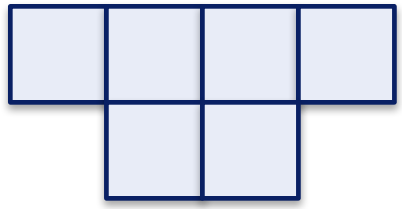
.....

Question 4

(2 marks)

Will this net fold to make a cube?
Give a reason.

Diagram NOT accurately drawn



.....

Exam Practice — Sketch a Net

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

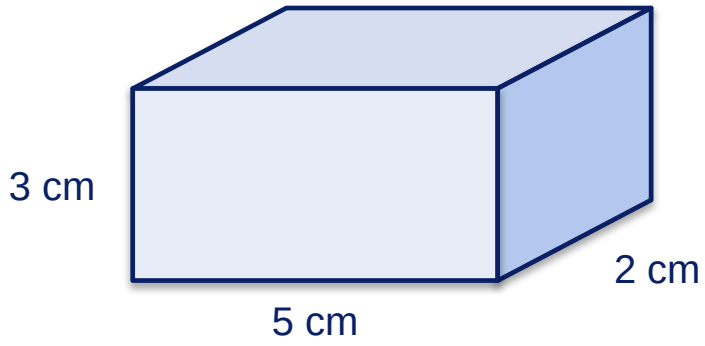
Hint Opposite faces are equal F + V − E = 2

Question 5

(3 marks)

Sketch a net of this cuboid.
Label the lengths.

Diagram NOT accurately drawn



.....

Question 6

(3 marks)

A prism has a pentagon at each end.

Work out the number of faces, edges and vertices.

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) 6 — B1

(b) 12 edges, 8 vertices — B2

B1 for one correct

Tip: count hidden edges too.

Question 2

(a) square-based pyramid — B1

(b) 5 — B1

Tip: 4 base corners and the top.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

(a) triangular prism B1

(b) 9 B1

Tip: 3 + 3 + 3 = 9 edges.

Question 4

No B1

Two faces would overlap B1

Tip: imagine folding it.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

6 rectangles that fold up	B1
pairs: 5 × 3, 5 × 2, 3 × 2	B1
lengths labelled	B1

Tip: opposite faces are the same.

Question 6

Faces	**= 5 + 2 = 7**	B1
Edges	**= 5 × 3 = 15**	B1
Vertices	**= 5 × 2 = 10**	B1

Tip: check: 7 + 10 − 15 = 2.